

WHAT CAN TORSION TELL US ABOUT REDUCTION TYPES ON ELLIPTIC CURVES?

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1. KODAIRA TYPES

Let K be a non-archimedean local field, and let $E := y^2 = x^3 + Ax + B$ be an elliptic curve (which is a smooth, projective, algebraic curve of genus one, on which there is a specified point $\mathcal{O}$), with coefficients in K . We want the equation to be *non-singular*, which means that the curve has no cusps or self-intersections. This condition is equivalent to the condition $\Delta := -16(4A^3 + 27B^2) \neq 0$, that is, being square-free in x . Δ is called the discriminant. Then, one can naïvely reduce this curve to get $y^2 = x^3 + \bar{a}x + \bar{b}$. This reduced equation may or may not be well-defined as it depends on the choice of the equation.

Example 1.1. For example $\{y^2 = x^3 + p^6\} \cong \{y^2 = x^3 + 1\}/\mathbb{Q}_p$ (the isomorphism is given by $x' = p^2x, y' = p^3y$), but one reduces to $y^2 = x^3$ and the other to $y^2 = x^3 + 1$. The reduction of the first equation is singular, but the elliptic curve does not have bad reduction. This singular reduction comes from a non-minimal equation, while the elliptic curve has another model with smooth reduction.

To resolve this ambiguity, we work with models. From this point until the end of the notes, we consider a non-archimedean local field K with ring of integers $\mathcal{O}_K$, and residue field k .

Example 1.2. Consider

$$E_n : y^2 = x^3 + x^2 + p^n \quad n \geq 1.$$

The same integral equation describes both a curve over $\mathbb{Q}_p$ and, after reducing modulo p , a curve over $\mathbb{F}_p$. Call

- the integral geometric object a *model*,
- its reduction modulo p the special *fibre*.

Here every E_n has the same special fibre in this model: $y^2 = x^2(x+1)$. These reduced equations are identical, but are the degenerations really identical?

The integral model is a two-dimensional geometric object, so an arithmetic surface, which can have singularities of its own. Resolving those singularities can introduce additional curves in the special fibre, while leaving the original elliptic curve over $\mathbb{Q}_p$ unchanged.

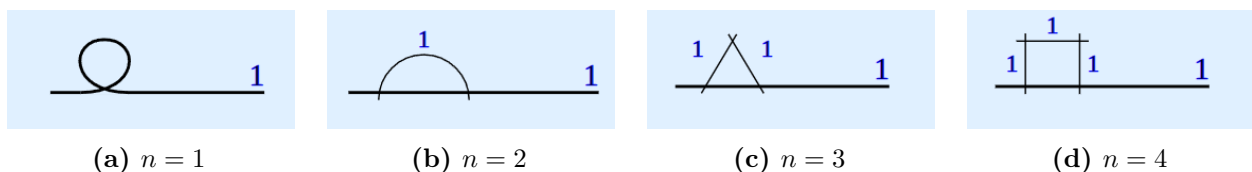


Figure 1. Kodaira types for $y^2 = x^3 + x^2 + p^n$

The pictures depict the special fibres after passing to the minimal regular proper model.

- Regular: the total arithmetic surface has no singularities, although its special fibre may still be singular.

- Minimal: no further component can be contracted while retaining a regular proper model of the same elliptic curve.

A unique such minimal model exists, up to the appropriate isomorphism. Each drawn component represents a curve, crossings record intersections, and labels record multiplicities.

One can see that each value of n gives a different minimal regular model. One can also see that the discriminant for each E_n is given by

$$\Delta(E_n) = -16p^n(4 + 27p^n), \quad v_p(\Delta(E_n)) = n.$$

So, reducing modulo p has erased the exponent n . But the elliptic curve over $\mathbb{Q}_p$ still remembers it. We would like a geometric description of the information that has been lost.

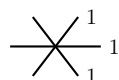
Definition 1.3. The Kodaira type records the configuration of the geometric special fibre of the minimal regular proper model, including how its components meet and their multiplicities.

The family $E_n := y^2 = x^3 + x^2 + p^n$ has Kodaira type I_n for each n . There are seven 10 possible Kodaira types (assuming $\text{char}(k) \neq 2, 3$):

$$I_0, I_0^*, I_n, I_n^*, II, II^*, III, III^*, IV, IV^*$$

Sadly, the special fibres of these minimal regular models can still be quite complicated.

Example 1.4. The special fibre of the minimal regular model of the curve $y^2 = x^3 + p^2$ is:



which has an ordinary triple point.

Definition 1.5. A curve over k is *semistable* if it is geometrically reduced and its only geometric singularities are ordinary double points. E/K has semistable reduction when it has a proper flat model with semistable special fibre.

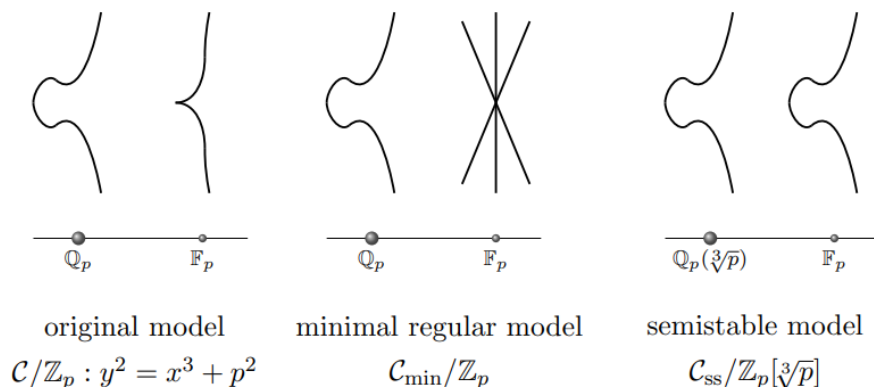
By contrast, semistable curves are all quite simple, however it is not always the case that a given curve has a semistable model. However,

Theorem 1.6 (Semistable Reduction Theorem). *There is always a finite extension K'/K where E achieves semistable reduction.*

Example 1.7. Let $E : y^2 = x^3 + p^2/\mathbb{Q}_p$. Set $u^3 = p$. Over $\mathbb{Q}_p(u)$, the substitution

$$x = u^2X, \quad y = u^3Y$$

gives $Y^2 = X^3 + 1$ which has good reduction for $p \geq 5$. Then,



If K'/K sufficiently large, the reduction type of E/K' is either good or multiplicative (I_0 or I_n). In this first case, we say E has *potentially good reduction*. In the second, it has *potentially multiplicative reduction*.

In terms of Kodaira types (assuming $\text{char}(k) \neq 2, 3$):

$$\underbrace{I_n, I_n^*}_{\text{(potentially) multiplicative}} \quad \underbrace{I_0, I_0^*, II, II^*, III, III^*, IV, IV^*}_{\text{(potentially) good}}$$

Given an explicit equation, one can use Tate's algorithm to compute the minimal regular model and obtain the Kodaira type.

2. REPRESENTATION THEORY

Turns out, representation theory can tell us a surprising amount about the geometry of the curve. Let $\ell \neq p$ be a prime.

Let K be a p -adic field, let I_K be inertia, and take $\ell \neq p$. Let $T_\ell(E) = \varprojlim_n E[\ell^n]$ be the Tate-module.

Write

$$\rho_\ell : I_K \longrightarrow \text{GL}_2(\mathbb{Z}_\ell)$$

for the action on $T_\ell(E)$. The classical criteria (Néron-Ogg-Shafarevich, Serre-Tate, Grothendieck) give:

Property of E/K	Condition on the inertia action
Good reduction	$\rho_\ell(I_K) = \{1\}$
Potentially good reduction	$\rho_\ell(I_K)$ is finite
Potentially multiplicative reduction	$\rho_\ell(I_K)$ is infinite

Once we know that we have potentially good reduction since the image of inertia is finite, we can recover the Kodaira type up to some ambiguities from the order of the image, if we assume $\text{char}(k) \neq 2, 3$,

Kodaira type	$\#(\rho(I_K))$
I_0	1
II, II^*	6
III, III^*	4
IV, IV^*	3
I_0^*	2

If we have potentially multiplicative reduction, we cannot necessarily recover the n in I_n .

Example 2.1. Consider the elliptic curves

$$E_1 : y^2 = x^3 + x^2 + 7 \quad E_2 : y^2 = x^3 + x^2 + 49,$$

over $\mathbb{Q}_7$. The Kodaira type of E_1 is I_1 and Kodaira type of E_2 is I_2 . Let $\ell = 5$. Under a suitable basis, we can write the images of inertia for Kodaira type I_n as

$$\rho_\ell(\sigma) = \begin{pmatrix} 1 & nt_\ell(\sigma) \\ 0 & 1 \end{pmatrix}, \quad (\sigma \in I_K)$$

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where t_ℓ is a fixed tame inertia character. So, we have

$$\rho_{E_1}(\sigma) = \begin{pmatrix} 1 & t_5(\sigma) \\ 0 & 1 \end{pmatrix}, \quad \rho_{E_2}(\sigma) = \begin{pmatrix} 1 & 2t_5(\sigma) \\ 0 & 1 \end{pmatrix}.$$

Now set

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}.$$

Because 2 is invertible in $\mathbb{Z}_5$, this is an invertible integral matrix. Direct multiplication gives

$$A\rho_{E_1}(\sigma)A^{-1} = \rho_{E_2}(\sigma) \quad \text{for every } \sigma \in I_K.$$

That is exactly what it means for the two inertia representations to be isomorphic:

$$T_5(E_1)|_{I_K} \cong T_5(E_2)|_{I_K}.$$

Thus, it is not possible to recover the n for multiplicative reduction from just the inertia action on the integral Tate module.

Alright. So, we can make a decent amount of progress with the entire integral Tate module, which is a compatible system of torsion points of all ℓ -power orders. What happens if we just restrict to just one layer of torsion, i.e., just $E[\ell]$?

Example 2.2. Consider:

$$E_1 : y^2 + y = x^3 \quad E_2 : y^2 + 3xy - 7y = x^3.$$

Over $\mathbb{Q}_7$, E_1 has good reduction (Kodaira type I_0) at 7 and E_2 has multiplicative reduction of Kodaira type I_3 .

From Neron-Ogg-Shafarevich, we know that E_1 must have trivial action by inertia.

Now, we compute $E_2[3](\overline{\mathbb{Q}}_7)$ and obtain that $E_2[3] = \langle (0, 0), (-7, 14 + 7\sqrt{-3}) \rangle$. Interestingly, $\sqrt{-3} \in \mathbb{Q}_7$ by Hensel's lemma, since $2^2 \equiv -3 \pmod{7}$, which means that $G_{\mathbb{Q}_7}$ acts trivially, and so $\bar{\rho}_{E_2, 3}(I_K) = \{1\}$.

Thus, at the level of $E[\ell]$ we cannot distinguish potentially good for potentially multiplicative reduction either.

Turns out, adding a little extra information, i.e, the p -adic distance between x -coordinates solves the problem.

Define for $\ell \neq 2, p$:

$$\mathcal{X}_\ell := \{x(P) : P \in E[\ell](\overline{K}) \setminus \mathcal{O}_E\}$$

and endow it with a distance function $d(x_1, x_2) = v_K(x_1 - x_2)$.

Theorem 2.3 (P., 2026). *Let K be a non-archimedean local field of residue characteristic p . Let E/K be an elliptic curve given by a Weierstrass equation.*

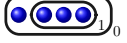
- *For any prime p , the points in $\mathcal{X}_\ell$ are equidistant if and only if the curve has potentially good reduction.*
- *For $p \geq 5$, the distances between points in $\mathcal{X}_\ell$ determine the Kodaira type of the special fibre. For $p = 2, 3$, assuming that E does not have potentially good reduction of type I_n^* ($n > 0$) (which can occur only when $p = 2$), the distances along with the inertia module $E[\ell]$ determine the Kodaira type.*

3. ℓ -TORSION CLUSTER PICTURES

We can encode these distance in the form of a picture. First, we note that to any elliptic curve equation, there is an associated ℓ -division polynomial ψ_ℓ , whose roots give you the x -coordinates of the ℓ -torsion points.

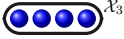
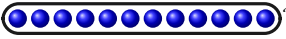
Example 3.1. Let us go back to $E : y^2 + 3xy - 7y = x^3$ over $\mathbb{Q}_7$ and let $\ell = 3$. We know that the x -coordinates of its 3-torsion points are given by

$$x_1 := 0, \quad x_2 := -7, \quad x_3 := 2 + \sqrt{-3}, \quad x_4 := 2 - \sqrt{-3}.$$

Let $M = \mathbb{Q}_7(E[3])/\mathbb{Q}_7$. One can check that $M = \mathbb{Q}_7$. A direct computation shows that $v_M(x_3 - x_i) = 0$ for $i = 1, 2, 4$, while $v_M(x_i - x_j) = 1$ for all distinct $i, j \in \{1, 2, 4\}$. Thus, x_1, x_2 , and x_4 are pairwise closer to each other than any of them is to x_3 . To compute the distances over $\mathbb{Q}_7$, we divide the valuations by $e_{M/\mathbb{Q}_7}$, which is 1. One can represent this pictorially via a cluster picture as ₀. Then, from the Theorem above, we see that we can recover n by noting that $d_0 = 0, d_1 = 1, \ell = 3$. Therefore $n = \ell\delta_1 = 3$, so the curve has Kodaira type I_3 .

In general, there are (unsurprisingly) two types of ℓ -torsion cluster pictures that occur:

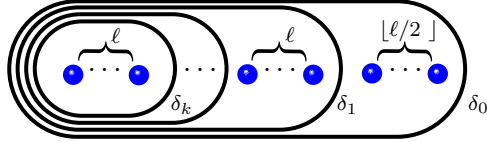
- (1) *Good cluster picture:* All distinct x -coordinates in $\mathcal{X}_\ell$ are equidistant from one another. In this case, $\mathcal{X}_\ell$ itself is the only cluster of size > 1 .

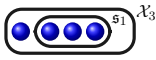
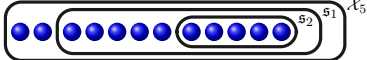
Example 3.2. $\ell = 3$:  $\ell = 5$: 

- (2) *Multiplicative cluster picture:* The x -coordinates partition into a strictly nested sequence of $k = \lfloor \ell/2 \rfloor$ proper subclusters. There exists a single chain of strictly contained clusters:

$$\mathfrak{s}_k \subsetneq \mathfrak{s}_{k-1} \subsetneq \cdots \subsetneq \mathfrak{s}_1 \subsetneq \mathcal{X}_\ell$$

where the innermost cluster has size $|\mathfrak{s}_k| = \ell$, and each subsequent layer contains ℓ additional points, such that $|\mathfrak{s}_i \setminus \mathfrak{s}_{i+1}| = \ell$ for $1 \leq i < k$. The complement in the maximal cluster contains $|\mathcal{X}_\ell \setminus \mathfrak{s}_1| = \lfloor \ell/2 \rfloor$ points.



Example 3.3. $\ell = 3$:  $\ell = 5$: 

Assume $p \geq 5$. For potentially good reduction, if the depth is d , then $6d \bmod 12$ recovers the valuation of the minimal discriminant and hence the Kodaira type. For potentially multiplicative reduction, looking at the parity of the outermost depth determines I_n and I_n^* , while the relative depth of the clusters recovers n .

4. PRIMES $p = 2, 3$

The reason things break down here is because the ℓ -torsion cluster pictures essentially recover the valuation of the minimal discriminant for potentially good reduction, and for $p \geq 5$, there is a one-to-one correspondence between Kodaira types and valuation of the minimal discriminant for potentially good reduction. This correspondence no longer holds for primes $p = 2, 3$. Not all hope is lost though, as for $p = 3$, one can recover the Kodaira type from the ℓ -torsion cluster picture along with the valuation of the conductor (which can be obtained from inertia), and then applying Ogg's formula.

For $p = 2$, the relationship between Kodaira type and potential reduction changes, as one can have potentially good reduction with Kodaira type I_n^* ($n > 0$). If we assume we don't have that special fibre, for potentially good, we can use the same trick as $p = 3$ of finding the valuation of the conductor and applying Ogg's formula. Once we know the curve has potentially multiplicative reduction, to distinguish between I_n vs I_n^* , we consider the residual Galois representation $\bar{\rho} : I_K \rightarrow \mathrm{GL}_2(\mathbb{F}_\ell)$ and

note that for I_n , $\bar{\rho}(I_K)$ must consist of only unipotent matrices, while I_n^* must have at least one non-unipotent matrix. Then, to recover the n in I_n^* , we need an additional quantity: $s_{L/K}$, which is the valuation of the different of a relevant quadratic ramified extension (can be obtained from inertial data).

There is no additional inertia information that would work for $p = 2$, by the way, as elliptic curves given by LMFDB labels 100048.g1 and 100048.n1 over $\mathbb{Q}_2$, have reduction types I_4^* and II^* respectively, but also have identical ℓ -torsion cluster pictures and for $\ell \neq 2$ have isomorphic Tate modules over $\mathbb{Q}_2^{unr}$. Thus, adding further information about the inertia action alone cannot distinguish this pair.